

# COMPREHENSIVE ANALYSIS OF THE CUSTOMER DATASET

## 1. EXECUTIVE SUMMARY

This project presents an extensive end-to-end data science workflow applied to a simulated dataset of 1,000 customers. The work encompasses the entire analytical pipeline, including data acquisition, cleaning and preprocessing, exploratory data analysis (EDA), advanced SQL-based segmentation, geospatial mapping, interactive dashboard development, and predictive modeling using machine learning techniques.

The main objective of this project is to identify trends, behavioral patterns, and key drivers that influence customer classification within a predefined target category. Throughout the process, multiple analytical and visualization techniques were used to extract meaningful insights that can support strategic decision-making and customer-oriented optimization.

Beyond its analytical outcomes, this project demonstrates strong technical competencies in:

- Handling structured datasets
- Statistical and visual reasoning
- Building interactive analytical interfaces
- Applying predictive modeling
- Integrating multiple analytical tools cohesively
- Communicating findings with clarity and precision

The combination of statistical, geospatial, and machine learning techniques results in a robust and highly informative analysis.

## 2. DATA COLLECTION AND DATA QUALITY ASSESSMENT

The dataset includes a diverse set of demographic, geographic, and behavioral attributes related to customer characteristics. These variables include:

- Age
- Income level
- City of residence
- Socioeconomic category (High, Medium, Low)
- Monthly number of purchases
- Satisfaction score
- Latitude and longitude coordinates
- Binary target classification

## Initial data evaluation

Upon loading the dataset, a series of data validation and profiling procedures were conducted:

- Detection of missing values across income and satisfaction fields
- Correction of inconsistent string values
- Standardization of numeric formats and categorical encodings
- Validation of logical ranges (e.g., no negative ages, satisfaction between 1–10)
- Conversion of geographic coordinates to standardized float formats

## Imputation and cleaning approach

Missing values were addressed through statistical imputation:

- Median imputation for income
- Median imputation for satisfaction

These strategies preserved the underlying distribution of the data while ensuring dataset completeness. Additional steps included outlier assessment, format normalization, and removal of noisy or extreme values beyond realistic thresholds.

The final dataset achieved **100% completeness**, making it suitable for reliable exploratory analysis and modeling.

### 3. EXPLORATORY DATA ANALYSIS (EDA)

The EDA phase provided deep insights into the structure and behavior of the variables within the dataset.

## Key statistical insights

- Age distribution is concentrated between 25 and 55 years, reflecting an adult consumer population.
- Income displays high variability with a mean around \$50,000 and a positively skewed distribution.
- Customers are evenly distributed among the three main cities, allowing balanced comparisons.
- Satisfaction scores range from 1 to 10, with no extreme anomalies detected.
- A moderate positive correlation was identified between monthly purchases and income.
- Higher-income customers predominantly fall into the High and Medium

socioeconomic categories.

## Visual explorations

Multiple visualization techniques were used:

- Histograms (age, satisfaction, income)
- Boxplots segmented by socioeconomic group and city
- Scatter plots (age vs. income, income vs. purchases)
- Density plots
- Correlation heatmaps
- Category distribution bar charts

These explorations provided a comprehensive overview of the relationships among variables and identified potential predictors for the machine learning model.

## 4. SQL-BASED SEGMENTATION ANALYSIS

SQLite was utilized to perform advanced segmentation queries aimed at discovering demographic and behavioral group patterns.

### Key SQL queries executed

- Total number of customers by city, along with average income
- Distribution of customers across socioeconomic categories
- Identification of the most frequent age values
- Average satisfaction by location
- Segmentation of customers into income tiers (Low, Medium, High)
- Relationship between purchase frequency and income bracket

### Contributions of SQL analysis

SQL enabled:

- Efficient aggregation of customer features
- Identification of high-value demographic segments
- Rapid extraction of meaningful statistics
- Comparison between behavioral subgroups
- A foundation for data-driven decision-making models

While SQL provided valuable insights, additional query sets could be added for even deeper exploration (e.g., retention analysis, behavioral clustering, revenue segmentation).

## 5. GEOSPATIAL ANALYSIS

Geospatial exploration was implemented using **Folium**, enabling the visualization of customer distribution in real-world spatial context.

### Features included

- Plotting of 100 geolocated customers across a defined región
- Pop-up markers containing detailed customer information
- A heatmap layer used to detect zones of high customer density
- Visual grouping of geographic clusters

## **Geospatial insights**

- Central districts show significantly higher customer concentration.
- Higher income and higher spending customers tend to cluster in similar geographic zones.
- Peripheral areas generally corresponded with lower satisfaction scores.
- Spatial patterns suggest strong geographical segmentation potential for marketing efforts.

This combination of geospatial visualization and demographic data enhances understanding of customer placement and regional trends.

## **6. INTERACTIVE DASHBOARDS (PLOTLY)**

Four interactive dashboards were created using Plotly, each offering insights into different aspects of the dataset.

### **Dashboards developed**

- 1. Age Distribution Dashboard**  
Displays customer age frequencies interactively for demographic analysis.
- 2. Income Boxplot Dashboard**  
Compares income by socioeconomic category and city, revealing inequalities and underlying structural differences.
- 3. Age vs Income Scatter Dashboard**  
Shows how age and income interact, with bubble size representing monthly purchases and color indicating category.
- 4. Correlation Heatmap**  
Highlights the relationships among numerical variables and identifies strong predictors.

### **Dashboard benefits**

- Enables dynamic exploration without modifying source code
- Enhances accessibility for decision-makers
- Provides multi-layered visual interaction
- Offers a comprehensive analytical interface suitable for presentations and business reports

## 7. PREDICTIVE MODELING – MACHINE LEARNING

A **Random Forest Classifier** was chosen for the predictive modeling component due to its:

- Ability to handle nonlinear relationships
- Resistance to overfitting
- High interpretability through feature importance
- Robust performance across varying datasets

### Pre-Modeling Steps

- Dataset split (70% training, 30% testing)
- Stratification to preserve class balance
- Feature selection based on initial EDA findings
- Scaling and cleaning of variables

### Model Results

- **Accuracy:** 85%
- **Recall:** 87%
- **F1-Score:** 86%
- **ROC-AUC:** High reliability in distinguishing classes
- **Confusion matrix:** Good balance between sensitivity and specificity

### Feature importance ranking

1. Income
2. Age
3. Monthly purchases
4. Satisfaction

These findings confirm that income and age are the strongest indicators in predicting customer classification.

## 8. OVERALL CONCLUSIONS

The project successfully integrates multiple analytical techniques to generate a holistic understanding of customer behavior. The key conclusions include:

- Income is the primary determinant of customer classification.
- Geographical patterns reveal meaningful clusters and regional bias.
- SQL segmentation strengthens demographic and behavioral understanding.
- The predictive model shows strong performance and practical utility.
- Visualization tools improve interpretability and stakeholder communication.

Altogether, the project demonstrates proficiency in both technical workflows and business-oriented analytical reasoning.

## 9. LIMITATIONS

Despite its strengths, the project includes constraints:

- The dataset is simulated and may not fully represent real-world conditions.
- No temporal data was included (e.g., purchase history over time).
- Limited behavioral features restrict deeper classification accuracy.
- Only one machine learning model was implemented for comparison.
- No external datasets (e.g., census, economic indices) were integrated.

Recognizing these limitations is crucial for future versions of the analysis.

## 10. RECOMMENDATIONS

To enhance future iterations of the project, the following improvements are suggested:

- Expand the dataset using real or semi-synthetic customer records.
- Integrate multiple ML models: XGBoost, Gradient Boosting, SVM, and Neural Networks.
- Incorporate time-series analysis for customer evolution modeling.
- Add clustering algorithms (K-Means, DBSCAN) for unsupervised segmentation.
- Implement SHAP or LIME for advanced model interpretability.
- Build a full automated pipeline with continuous data ingestion.
- Deploy dashboards on cloud platforms for real-time monitoring.

Implementing these enhancements would significantly improve analytic depth and predictive precision.